Synoptic Question 15: The Physics of Fusion Energy

Reading Passage

Read the following article and then answer the questions that follow.

Nuclear fusion is the process that powers the Sun and other stars. It involves light atomic nuclei combining to form a heavier nucleus, releasing an immense amount of energy in the process. Harnessing this on Earth as a clean, safe, and virtually limitless energy source is one of the greatest scientific and engineering challenges.

The most promising reaction for a first-generation fusion power plant is the "D-T" reaction, where a deuterium nucleus (^2_1H) fuses with a tritium nucleus (^3_1H). This process releases energy according to Einstein's mass-energy equivalence principle, $E = mc^2$, because the total mass of the products is less than the total mass of the reactants. This "lost mass" is converted into kinetic energy.

To overcome the strong electrostatic repulsion between the positively charged nuclei, the D-T fuel must be heated to extreme temperatures—over 150 million Kelvin. At this temperature, the fuel is no longer a gas but a plasma: a "soup" of free-moving positive ions and negative electrons.

Since no solid material can withstand this temperature, the hot plasma must be confined without touching the walls of the reactor. The most common method is magnetic confinement, used in a donut-shaped device called a "tokamak". A powerful magnetic field forces the charged ions and electrons to spiral around the magnetic field lines, trapping them in the centre of the vessel. The goal is to sustain this hot, dense plasma for long enough for a net energy gain to be achieved.

Data:

Data for the D-T fusion reaction and a hypothetical tokamak:

Table 1: Data for Fusion Analysis

Parameter	**Value**
Mass of Deuterium (^2_1H) nucleus	3.3436×10^{-27} kg
Mass of Tritium (^3_1H) nucleus	5.0083×10^{-27} kg
Mass of Helium (^4_2He) nucleus	6.6465×10^{-27} kg
Mass of a neutron, m_n	1.6749×10^{-27} kg
Speed of light, c	3.00×10^8 m s^{-1}
Charge of a proton, e	1.60×10^{-19} C
Boltzmann constant, k_B	1.38×10^{-23} J K^{-1}
Typical plasma temperature, T	1.50×10^8 K
Magnetic field in tokamak, B	5.0 T
Molar mass of Deuterium	2.0 g mol^{-1}
Molar mass of Tritium	3.0 g mol^{-1}
Avogadro's constant, N_A	6.02×10^{23} mol^{-1}

Questions

(a) The D-T fusion reaction produces a helium nucleus and a neutron.

- (i) Write the balanced nuclear equation for this reaction.
- (ii) Calculate the mass defect (Δm) for this reaction.
- (iii) Calculate the energy released (in joules) from a single D-T reaction.

(b) The energy scale of this reaction.

- (i) Convert the energy released per reaction into mega-electron volts (MeV).
- (ii) Calculate the total energy released (in joules) from the fusion of 2.0 g of deuterium with 3.0 g of tritium (a 1:1 molar mix).

(c) The plasma must be at an extremely high temperature.

- (i) Explain in terms of forces why such high temperatures are required to initiate fusion.

(ii) Assuming the ions in the plasma have an average thermal energy given by $E_K = \frac{3}{2}k_BT$, calculate the average kinetic energy (in joules) of a D-T ion at 150 million K.

(d) A hypothetical power plant is designed to produce a thermal power output of 1200 MW.

(i) Calculate the number of D-T reactions that must occur per second to produce this power.

(ii) This thermal power is used to boil water and run a steam turbine, which converts the thermal energy into electrical energy with an efficiency of 35%. Calculate the electrical power output of the plant.

(e) The plasma is confined by a strong magnetic field.

(i) Explain what a plasma is and why a magnetic field is able to confine it.

(ii) A tritium ion (^3_1H) is travelling at 6.0×10^6 m s^{-1} perpendicular to the 5.0 T magnetic field. Calculate the magnitude of the magnetic force acting on it.

(iii) This force causes the ion to move in a circular path. Calculate the radius of this path.

(f) State one significant challenge, other than confinement, that must be overcome to build a practical fusion power plant.

Solutions

(a) (i) ${}^2_1H + {}^3_1H \rightarrow {}^4_2He + 1_0n$

(ii) Mass of reactants $= m_D + m_T = (3.3436 \times 10^{-27}) + (5.0083 \times 10^{-27}) = 8.3519 \times 10^{-27}$ kg.
Mass of products $= m_{He} + m_n = (6.6465 \times 10^{-27}) + (1.6749 \times 10^{-27}) = 8.3214 \times 10^{-27}$ kg.
Mass defect $\Delta m = \text{Mass}_{reactants} - \text{Mass}_{products} = (8.3519 - 8.3214) \times 10^{-27} = 3.05 \times 10^{-29}$ kg.

(iii) Energy $E = \Delta mc^2 = (3.05 \times 10^{-29}) \times (3.00 \times 10^8)^2 = 2.745 \times 10^{-12}$ J.

(b) (i) $E = (2.745 \times 10^{-12}\ \text{J})/(1.60 \times 10^{-19}\ \text{J/eV}) = 1.716 \times 10^7$ eV $= 17.16$ MeV.

(ii) 2.0 g of Deuterium is 1.0 mole. 3.0 g of Tritium is 1.0 mole.
The mix contains 1.0 mole of D-T pairs, so N_A reactions will occur.
Total Energy = (Energy per reaction) $\times N_A$
Total Energy $= (2.745 \times 10^{-12}) \times (6.02 \times 10^{23}) = 1.65 \times 10^{12}$ J.

(c) (i) The deuterium and tritium nuclei are both positively charged. They experience a strong electrostatic (Coulombic) repulsion. To fuse, they must get extremely close (to within the range of the strong nuclear force). High temperatures give the nuclei extremely high kinetic energy, allowing them to overcome this repulsive force.

(ii) $E_K = \frac{3}{2}k_BT = \frac{3}{2}(1.38 \times 10^{-23})(1.50 \times 10^8) = 3.105 \times 10^{-15}$ J.

(d) (i) $P_{th} = 1200$ MW $= 1.20 \times 10^9$ W (J s^{-1}).
Reactions per second $= \frac{\text{Total Power}}{\text{Energy per reaction}} = \frac{1.20 \times 10^9}{2.745 \times 10^{-12}} = 4.37 \times 10^{20}$ s^{-1}.

(ii) Electrical Power = Thermal Power $\times$ Efficiency
$P_{elec} = (1.20 \times 10^9) \times 0.35 = 4.20 \times 10^8$ W $= 420$ MW.

(e) (i) A plasma is a state of matter where atoms are heated to such high temperatures that electrons are stripped from their nuclei, creating a mixture of free-moving positive ions and negative electrons. Because it consists of charged particles, its motion can be controlled by magnetic fields via the Lorentz force ($F = qvB$).

(ii) A tritium ion (${}^3_1H$) has one proton, so its charge $q = e = 1.60 \times 10^{-19}$ C.
Force $F = qvB\sin\theta$. Since it is perpendicular, $\sin\theta = 1$.
$F = (1.60 \times 10^{-19})(6.0 \times 10^6)(5.0) = 4.80 \times 10^{-12}$ N.

(iii) The magnetic force provides the centripetal force.
$F_c = F_{mag} \implies \frac{mv^2}{r} = qvB$
$r = \frac{mv}{qB} = \frac{(5.0083 \times 10^{-27})(6.0 \times 10^6)}{(1.60 \times 10^{-19})(5.0)} = 0.0376$ m (or 3.76 cm).

(f) Any one of the following:

- **Tritium Breeding:** Tritium is radioactive and rare. It must be "bred" within the reactor, e.g., by having the fusion neutrons strike a lithium blanket.
- **Materials Science:** Finding materials for the reactor walls that can withstand intense neutron bombardment without becoming overly radioactive or structurally weak.
- **Energy Loss:** Preventing the plasma from losing energy too quickly (e.g., via radiation) so that it remains hot enough for sustained fusion.

[Diagram of a Tokamak showing plasma confined by magnetic fields]

Figure 1: Magnetic Confinement in a Tokamak Reactor

Cambridge IGCSE Physics (0625)

Paper 4 Theory (Extended)

Synoptic Topic: Fusion Energy
Variant 2 (Higher Difficulty)

Time: 1 hour 15 minutes **Total Marks:** 50

Instructions

- Answer all questions.
- Use a black or dark blue pen. Show your working in calculations.
- Give answers to an appropriate number of significant figures.

Data

- Mass of deuterium nucleus, $m_D = 3.3436 \times 10^{-27}$ kg
- Mass of tritium nucleus, $m_T = 5.0083 \times 10^{-27}$ kg
- Mass of helium-4 nucleus, $m_{He} = 6.6465 \times 10^{-27}$ kg
- Mass of neutron, $m_n = 1.6749 \times 10^{-27}$ kg
- Speed of light, $c = 3.00 \times 10^8$ m s^{-1}
- Elementary charge, $e = 1.60 \times 10^{-19}$ C
- Boltzmann constant, $k_B = 1.38 \times 10^{-23}$ J K^{-1}
- Avogadro constant, $N_A = 6.02 \times 10^{23}$ mol^{-1}

Question 1: Fusion reaction physics (14 marks)

The principal fusion reaction considered for early power plants is

$$^{2}_{1}\text{H} + ^{3}_{1}\text{H} \rightarrow ^{4}_{2}\text{He} + ^{1}_{0}\text{n} + \text{energy}.$$

(a) State which fundamental interaction is responsible for binding nucleons when fusion occurs. [1]

(b) Calculate the mass defect, Δm, for one D–T reaction. [2]

(c) Hence calculate the energy released per reaction in:

  (i) joules [2]

  (ii) MeV. [2]

(d) A reactor produces thermal power of 1.50×10^9 W.

(i) Determine the required number of D–T reactions per second. [2]

(ii) Explain why this very large reaction rate does not imply that a large mass of fuel is consumed each second. [2]

(e) Open-ended:

> Some people claim fusion is "essentially limitless" and therefore fuel supply is not a practical concern. Discuss this statement using physics ideas, including at least one argument that supports the claim and at least one realistic limitation.

[3]

Question 2: Plasma and confinement (16 marks)

A tokamak confines a fully ionised D–T plasma at temperature $T = 1.50 \times 10^8$ K in a magnetic field of magnitude $B = 5.0$ T.

(a) Explain what is meant by a plasma. [2]

(b) Show that the average thermal kinetic energy per particle is approximately 3.1×10^{-15} J. [2]

(c) A tritium ion moves at $6.0 \times 10^6\ \mathrm{m\,s^{-1}}$ perpendicular to the magnetic field.

(i) Calculate the magnetic force on the ion. [2]

(ii) Determine the radius of its circular path. [3]

(d) The same ion instead moves at $6.0 \times 10^6\ \mathrm{m\,s^{-1}}$ at an angle of 30° to the magnetic field.

(i) State the shape of its path. [1]

(ii) Calculate the new radius of curvature. [2]

(e) Open-ended:

> Evaluate whether "increasing magnetic field strength is always the best way to improve confinement". Your answer should include at least two physics-based advantages and two practical limitations.

[4]

Question 3: Power generation and systems analysis (20 marks)

A fusion station delivers thermal power to a steam turbine-generator.

(a) The turbine-generator has efficiency $\eta = 0.38$.

(i) Calculate the electrical output power for thermal input 1.50×10^9 W. [2]

(ii) State two reasons why efficiency is less than 100%. [2]

(b) During operation, 18% of thermal power is transferred to seawater used for cooling. The cooling water temperature rise is $\Delta T = 10\,^{\circ}\mathrm{C}$. Take specific heat capacity of water as $4200\,\mathrm{J\,kg^{-1}\,^{\circ}C^{-1}}$.

(i) Calculate the rate of energy transfer to cooling water. [2]

(ii) Calculate the mass flow rate of cooling water required. [3]

(c) One design uses a lithium blanket to breed tritium through neutron interactions. Explain why tritium breeding is necessary for long-term operation. [3]

(d) Open-ended:

> A country is choosing between building one large fusion plant and many smaller renewable plants. Using physics arguments only (energy density, reliability, losses, conversion stages, and system response), discuss the key trade-offs and justify one strategy.

[8]

Sample Solutions – Variant 2

Question 1

(a) Strong nuclear interaction.

(b) $m_{\text{reactants}} = m_D + m_T = (3.3436 + 5.0083) \times 10^{-27} = 8.3519 \times 10^{-27}\,\text{kg}$
$m_{\text{products}} = m_{He} + m_n = (6.6465 + 1.6749) \times 10^{-27} = 8.3214 \times 10^{-27}\,\text{kg}$
$\Delta m = 8.3519 \times 10^{-27} - 8.3214 \times 10^{-27} = 3.05 \times 10^{-29}\,\text{kg}$.

(c) (i) $E = \Delta mc^2 = (3.05 \times 10^{-29})(3.00 \times 10^8)^2 = 2.745 \times 10^{-12}\,\text{J}$.

(ii) $E = \dfrac{2.745 \times 10^{-12}}{1.60 \times 10^{-19}} = 1.72 \times 10^7\,\text{eV} = 17.2\,\text{MeV}$.

(d) (i) $N = \dfrac{P}{E} = \dfrac{1.50 \times 10^9}{2.745 \times 10^{-12}} = 5.46 \times 10^{20}\,\text{s}^{-1}$.

(ii) Each reaction involves only two nuclei, each with mass of order 10^{-27} kg, so even $\sim 10^{20}$ reactions s^{-1} corresponds to small fuel mass flow (mg scale per second).

(e) Indicative points: deuterium is abundant in seawater (supports "limitless"); tritium is scarce and must be bred; infrastructure/material constraints and energy economics limit practical availability.

Question 2

(a) Ionised gas of free electrons and positive ions with collective electromagnetic behavior.

(b) $E_k = \dfrac{3}{2}k_BT = \dfrac{3}{2}(1.38 \times 10^{-23})(1.50 \times 10^8) = 3.11 \times 10^{-15}\,\text{J}$.

(c) (i) $F = qvB = (1.60 \times 10^{-19})(6.0 \times 10^6)(5.0) = 4.8 \times 10^{-12}\,\text{N}$.

(ii) $r = \dfrac{mv}{qB} = \dfrac{(5.0083 \times 10^{-27})(6.0 \times 10^6)}{(1.60 \times 10^{-19})(5.0)} = 3.76 \times 10^{-2}\,\text{m}$.

(d) (i) Helical path.

(ii) Perpendicular component $v_\perp = v \sin 30^\circ = 3.0 \times 10^6\,\text{m s}^{-1}$, $r = \dfrac{mv_\perp}{qB} = 1.88 \times 10^{-2}\,\text{m}$.

(e) Indicative discussion: higher B increases Lorentz force and reduces gyro-radius, aiding confinement; but coil stresses, superconducting limits, cost, plasma instabilities, and engineering complexity constrain gains.

Question 3

(a) (i) $P_{elec} = \eta P_{th} = 0.38(1.50 \times 10^9) = 5.70 \times 10^8\,\text{W}$.

(ii) Any two: friction/turbulence, finite temperature differences, electrical resistance, exhaust heat losses.

(b) (i) $P_{cool} = 0.18(1.50 \times 10^9) = 2.70 \times 10^8$ W.

(ii) $P = \dot{m}c\Delta T$ so $\dot{m} = \dfrac{2.70 \times 10^8}{4200 \times 10} = 6.43 \times 10^3$ kg s^{-1}.

(c) Tritium half-life is short and natural abundance is negligible; without breeding, external supply limits reactor uptime. Neutrons from D–T reactions can convert lithium into tritium to close fuel cycle.

(d) Indicative argument: large fusion offers high continuous power density and reduced storage need, but centralised losses and complex outages; distributed renewables reduce single-point failure and transmission risk but require storage/grid balancing due to intermittency. A justified strategy with physics evidence gains full credit.

Mark Scheme – Variant 2 (CIE P4 style)

General marking principles

Award marks for valid physics reasoning and correct application of formulae. Allow ecf (error carried forward) where method is correct.

Q1 (14)

- (a) strong nuclear force identified – 1
- (b) reactant mass correct – 1; product mass and subtraction for Δm correct – 1
- (c)(i) substitution into $E = mc^2$ – 1; correct value/unit – 1
- (c)(ii) correct conversion J to eV/MeV method – 1; correct value – 1
- (d)(i) uses $N = P/E$ – 1; correct rate – 1
- (d)(ii) explanation links tiny mass per nucleus to high count rate – 2
- (e) open-ended level mark: Level 3 (3): balanced discussion with support + limitation, clear physics links. Level 2 (2): some valid physics but partial balance/detail. Level 1 (1): limited statement with minimal physics.

Q2 (16)

- (a) plasma as ionised gas with free charges – 2
- (b) correct formula/substitution – 1; correct value – 1
- (c)(i) correct use of $F = qvB$ (or $qvB\sin\theta$) – 1; value/unit – 1
- (c)(ii) method equating magnetic and centripetal effect/use of $r = mv/(qB)$ – 2; value – 1
- (d)(i) helical path – 1
- (d)(ii) uses $v_\perp = v\sin 30^\circ$ – 1; correct radius – 1
- (e) open-ended level mark: Level 4 (4): two+ clear advantages and two+ realistic limitations with sound physics. Level 3 (3): good physics with one side less developed. Level 2 (2): basic but relevant points. Level 1 (1): limited statements, weak physics.

Q3 (20)

- (a)(i) method with efficiency relation – 1; correct power – 1
- (a)(ii) two acceptable loss mechanisms – 2
- (b)(i) applies fraction of thermal power – 1; correct value – 1
- (b)(ii) uses $P = \dot{m}c\Delta T$ rearranged – 2; correct flow rate – 1
- (c) need for breeding linked to scarcity/half-life and neutron-lithium process – 3
- (d) open-ended level mark: Level 4 (7–8): coherent comparative argument, multiple accurate physics trade-offs, justified conclusion. Level 3 (5–6): mostly accurate comparison with some justification. Level 2 (3–4): limited comparison with partial physics. Level 1 (1–2): simple statements, little physics linkage.

End of Mark Scheme